

Absolute Error and MAE

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Compute squared loss for Absolute and squared loss. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Absolute and squared loss. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Residuals $[-2, 0, 4]$ m: calculate MAE MSE RMSE and the result of converting to cm.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

6. Solve the given case

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Actual heights are 4, 8 and 10 cm; predictions are 5, 6 and 13 cm. Calculate absolute errors and MAE.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Compute MAE

Calculate MAE for the worked residuals.

My reasoning and evidence

13. W2 Compute squared loss

Give MSE and RMSE with units.

My reasoning and evidence

14. W3 Compare patterns

Why do $[0, 0, 6]$ and $[2, 2, 2]$ have equal MAE but unequal MSE?

My reasoning and evidence

15. W4 Apply independently

Residuals $[-2, 0, 4]$ m: calculate MAE MSE RMSE and the result of converting to cm.

My reasoning and evidence

16. W5 Choose honestly

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Absolute Error and MAE

Compare absolute and squared error without losing track of units.

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Absolute Error and MAE

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Absolute errors

Absolute values 1, 2, 3 give $MAE 6/3 = 2$ cm.



2. Squared errors

Squares 1, 4, 9 give $MSE 14/3$ cm², approximately = 4.67 cm².



3. Root squared error

$RMSE = \sqrt{14/3}$, approximately 2.16 cm.

CHANGE ONE THING

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Why is quoting MSE as 4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare absolute and squared error without losing track of units.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? $RMSE = \sqrt{14/3}$, approximately 2.16 cm.

1. Absolute errors
2. Root squared error
3. Squared errors

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Root squared error $RMSE = \sqrt{14/3}$, approximately 2.16 cm. Read the picture in order. $RMSE = \sqrt{14/3}$, approximately 2.16 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Convert every residual to millimetres. Which conclusion is supported by the supplied facts?

1. MAE becomes 20 mm; MSE becomes $1400/3 \text{ mm}^2$; RMSE becomes about 21.60 mm.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: MAE becomes 20 mm; MSE becomes 1400/3 mm²; RMSE becomes about 21.60 mm. Residuals scale by 10; squares scale by 100, while the square root returns a length unit.

Repair a missing building block

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on training targets. Short practice: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using evidence. Short practice: What evidence would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and prediction pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
